

# Hongjie Wu

PH.D. IN ENGINEERING · SICHUAN UNIVERSITY

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"Rome was not built in a day."

## Summary

I received my Ph.D. in Engineering from the Data Intelligence and Computing Art Lab, College of Computer Science, Sichuan University, China, under the guidance of Prof. Jiancheng Lv.

My research interests include generative models (more focus on diffusion and flow models), computer vision, efficient AI, multi-modal learning, and broad deep learning topics in general. I am a highly diligent and self-motivated research student.

## Education

### Sichuan University

PH.D. CANDIDATE IN SOFTWARE ENGINEERING

Chengdu, China

Sept. 2022 - Current

- **Supervisor:** Jiancheng Lv
- **Honor:** Top 10 Student Collectives, Ph.D. Innovation Scholarship, NPIC Scholarship, The Star of Graduate Students, Outstanding Ph.D. Student of Sichuan University

### Sichuan University

MASTER IN COMPUTER TECHNOLOGY

Chengdu, China

Sept. 2019 - Jun. 2022

- **Supervisor:** Jiancheng Lv
- **Honor:** Outstanding Graduates of Sichuan Province, The Academic Star of Graduate Students, First-class University Scholarship, Outstanding Graduates of Sichuan University

### Sichuan University

BACHELOR IN COMPUTER SCIENCE AND TECHNOLOGY

Chengdu, China

Sept. 2014 - Jun. 2018

- **Supervisor:** Lei Zhang

## Publications/ Manuscripts

### RESEARCH AREA I: DIFFUSION MODEL

#### Enhancing Diffusion Model Stability for Image Restoration via Gradient Management

Hongjie Wu, Mingqin Zhang, Linchao He, Ji-Zhe Zhou, Jiancheng Lv

Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM) Oral presentation, 2025

#### ICCR-Diff: Identity-Preserving and Controllable Craniofacial Reconstruction with Diffusion Models

Mingqin Zhang, Hongjie Wu, Zhengqing Zang, Jian Wang, Chaoqun Niu, Yuan Li, Jiancheng Lv

Knowledge-Based Systems p. 114554, 2025

#### Diffusion Posterior Proximal Sampling for Image Restoration

Hongjie Wu\*, Linchao He\*, Mingqin Zhang, Dongdong Chen, Kunming Luo, Mengting Luo, Ji-Zhe Zhou, Hu Chen, Jiancheng Lv

Proceedings of the 32st ACM International Conference on Multimedia (ACM MM) Oral presentation, 2024

#### Fast and Stable Diffusion Inverse Solver with History Gradient Update

Linchao He, Hongyu Yan, Mengting Luo, Hongjie Wu, Kunming Luo, Wang Wang, Wenchao Du, Hu Chen, Hongyu Yang, Yi Zhang, Jiancheng Lv

arXiv preprint arXiv:2307.12070, 2023

### RESEARCH AREA II: MULTI-VIEW LEARNING

#### Euclidean Distance Is Not Your Swiss Army Knife

Yuze Tan, Yixi Liu, Hongjie Wu, Shudong Huang, Zenglin Xu, Ivor Tsang, Jiancheng Lv

IEEE Transactions on Knowledge and Data Engineering (TKDE), 2024

#### Preserving Local and Global Information: An Effective Metric-based Subspace Clustering

Yixi Liu, Yuze Tan, Hongjie Wu, Shudong Huang, Yazhou Ren, Jiancheng Lv

Proceedings of the 31st ACM International Conference on Multimedia (ACM MM), 2023

#### Metric multi-view graph clustering

Yuze Tan, Yixi Liu, Hongjie Wu, Jiancheng Lv, Shudong Huang

Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2023

#### Pure graph-guided multi-view subspace clustering

Hongjie Wu, Shudong Huang, Chenwei Tang, Yancheng Zhang, Jiancheng Lv

Multi-view subspace clustering on topological manifold

Shudong Huang\*, Hongjie Wu\*, Yazhou Ren, Ivor Tsang, Zenglin Xu, Wentao Feng, Jiancheng Lv  
Advances in Neural Information Processing Systems (NeurIPS) Spotlight presentation pp. 25883–25894. 2022

Multi-view graph clustering by adaptive manifold learning

Peng Zhao\*, Hongjie Wu\*, Shudong Huang  
Mathematics p. 1821. MDPI, 2022

RESEARCH AREA III: DEEP LEARNING APPLICATIONS

Self-supervised Domain Adaptation with Significance-Oriented Masking for Pelvic Organ Prolapse detection

Shichang Li, Hongjie Wu, Chenwei Tang, Dongdong Chen, Yueyue Chen, Ling Mei, Fan Yang, Jiancheng Lv  
Pattern Recognition Letters. 2024

ST-CA YOLOv5: Improved YOLOv5 based on Swin Transformer and coordinate attention for surface defect detection

Wen Yang, Hongjie Wu, Chenwei Tang, Jiancheng Lv  
International Joint Conference on Neural Networks (IJCNN), 2023

Automatic cataract detection with multi-task learning

Hongjie Wu, Jiancheng Lv, Jian Wang  
International Joint Conference on Neural Networks (IJCNN), 2021

Honors & Awards

|      |                                                                                                  |       |
|------|--------------------------------------------------------------------------------------------------|-------|
| 2026 | Outstanding Graduate Students of Province, granted by Sichuan Provincial Department of Education | China |
| 2026 | Outstanding Graduates of University, granted by Sichuan University                               | China |
| 2025 | Ph.D. Innovation Scholarship, granted by Ministry of Education                                   | China |
| 2025 | Suzhou Industrial Park Scholarship, granted by Suzhou Industrial Park                            | China |
| 2024 | Top 10 Student Collectives (team leader), granted by Sichuan University                          | China |
| 2023 | Ph.D. Innovation Scholarship, granted by Ministry of Education                                   | China |
| 2023 | NPIC Scholarship (first-class), granted by NPIC                                                  | China |
| 2023 | Outstanding Ph.D. Student of Sichuan University, granted by Sichuan University                   | China |
| 2022 | The Academic Star of Graduate Students, granted by Sichuan University                            | China |
| 2022 | Outstanding Graduate Students of Province, granted by Sichuan Provincial Department of Education | China |
| 2021 | Outstanding Graduates of University, granted by Sichuan University                               | China |
| 2021 | First-class University Scholarship, granted by Ministry of Education                             | China |

Professional Services

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|-------|----------------------------------------------------------------------------------|
| 2026- | Reviewer, Neural Information Processing Systems (NeurIPS)                        |
| 2026- | Reviewer, European Conference on Computer Vision (ECCV)                          |
| 2025- | Reviewer, Computer Vision and Pattern Recognition (CVPR)                         |
| 2025- | Reviewer, IEEE Transactions on Artificial Intelligence (TAI)                     |
| 2025- | Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) |
| 2025- | Program Committee Member, AAAI Conference on Artificial Intelligence             |
| 2025- | Reviewer, The Journal of Supercomputing                                          |
| 2025- | Reviewer, IEEE Transactions on Image Processing (TIP)                            |
| 2025- | Reviewer, Knowledge-Based Systems (KBS)                                          |
| 2024- | Reviewer, Big Data Mining and Analytics                                          |
| 2024- | Reviewer, ACM International Conference on Multimedia (ACM MM)                    |
| 2024- | Reviewer, International Joint Conference on Neural Networks (IJCNN)              |
| 2023- | Reviewer, Journal of Computer Research and Development                           |
| 2023- | Reviewer, Pattern Recognition                                                    |

Interests

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|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| Running         | Running provides me with a profound sense of mental and physical well-being and contentment.                          |
| Reading         | I love reading books, for they ignite my imagination and broaden my knowledge and understanding of the world.         |
| Watching Movies | I like to watch movies in my free time, which have been a captivating source of entertainment and fascination for me. |
| Cooking         | I am good at cooking Sichuan cuisine (perhaps), and preparing delicious meals gives me a great sense of achievement.  |